# IMEN266 Operations Research II
# Chapter 5. Continuous-time Markov chains (CTMCs)

Fall, 2026




INDUSTRIAL AND MANAGEMENT
ENGINEERING, POSTECH

# Outline

1. Exponential distribution
2. Poisson process
3. Continuous-time Markov chains (CTMC)

# How to study this chapter (2026 format)

- **Slides** (this file): concepts, class-problem statements, and the worked derivations developed in class. No transcription needed — definitions, theorems and proofs also live in the Companion.
- **Notebooks** (CP6 exponential/Poisson · CP7 CTMC modeling · CP8 transient/limiting/cost): in class the `INCLASS` version (fill the `TODO`s, `check()` answers instantly); at home the full version with solutions and experiments.
- **Companion** (gap-fill $\rightarrow$ full): all definitions and theorems + 8 core proofs with blanks, checkpoints, AI study cards.
- **Homework 3**: A exam-style (no AI first) · B critique an AI solution · C stress-test the exponential assumption · D prompt log.
- Class rhythm (75 min): quiz $5'$ $\rightarrow$ concepts $\rightarrow$ live notebook $\rightarrow$ wrap-up. Ch. 5–6 = weeks 6–8.
- Ground truth = analytic $\leftrightarrow$ simulation agreement, not AI confidence. Materials: PLMS and GitHub `imen266-2026`.

# Recap

- A random variable $X$ is exponentially distributed with parameter $\lambda$ if its pdf is

$$f(x) = \begin{cases} \lambda e^{-\lambda x} & \text{if } x > 0 \\ 0 & \text{otherwise,} \end{cases}$$

and its cdf is

$$F(x) = \begin{cases} 1 - e^{-\lambda x} & \text{if } x > 0 \\ 0 & \text{otherwise.} \end{cases}$$

- The mean and variance are

$$\mathsf{E}[X] = \frac{1}{\lambda}$$

$$\mathsf{Var}[X] = \frac{1}{\lambda^2}.$$

▷ Companion §1.1: exponential distribution — pdf, cdf, mean, variance; memorylessness is Proof 1.

# Important properties

1. Memoryless property (always as good as new)

$$P\left(X > t + s | X > s\right) = P(X > t) = e^{-\lambda t}.$$

2. Minimum of independent exponentials

   Suppose $X_1 \sim \exp(\lambda_1), \ldots, X_n \sim \exp(\lambda_n)$.
   Define a new random variable

$$Z = \min(X_1, \ldots, X_n).$$

   Then, $Z \sim \exp(\lambda_1 + \cdots + \lambda_n)$.

▷ Companion Proof 1: memorylessness, the minimum of independent exponentials, and which clock rings first.

# Important properties

## 1. Memoryless: a used one is as good as a new one

$$P(X > t+s \mid X > s) = \frac{P(X > t+s,\ X > s)}{P(X > s)} = \frac{P(X > t+s)}{P(X > s)}$$

$$= \frac{e^{-\lambda(t+s)}}{e^{-\lambda s}} = e^{-\lambda t} = P(X > t)\,.$$

Read it with $s =$ the age of the "used one" and $t =$ the extra life asked of it: the residual life of a used component has the same $\exp(\lambda)$ distribution as the life of a new one.

## 2. Minimum of independent exponentials

$$P(Z > z) = P\big(\min(X_1, \ldots, X_n) > z\big) = P(\text{all } X_i > z)$$

$$= P(X_1 > z) \cdots P(X_n > z) \qquad \text{(independence)}$$

$$= e^{-\lambda_1 z} \cdots e^{-\lambda_n z} = e^{-(\lambda_1 + \cdots + \lambda_n) z}\,.$$

▷ So $Z \sim \exp(\lambda_1 + \cdots + \lambda_n)$: the rates add. Independence is the one assumption that cannot be dropped — "all $X_i > z$" factorises only then.

# Class problem 6

1. I receive emails such that the time between two new emails is distributed exponentially with mean 2 minutes. I received an email at 10 am.
   (a) What is the probability that I receive the next email before 10:04am?
   (b) If I did not receive any email till 10:05am, what is the probability that I will receive no emails till 10:07am?
   (c) If I did not receive any email till 10:10am, when can I expect the next email?

▶ Colab notebook: CP6 §1 (e-mails) `[INCLASS] [full]`

# Class problem 6

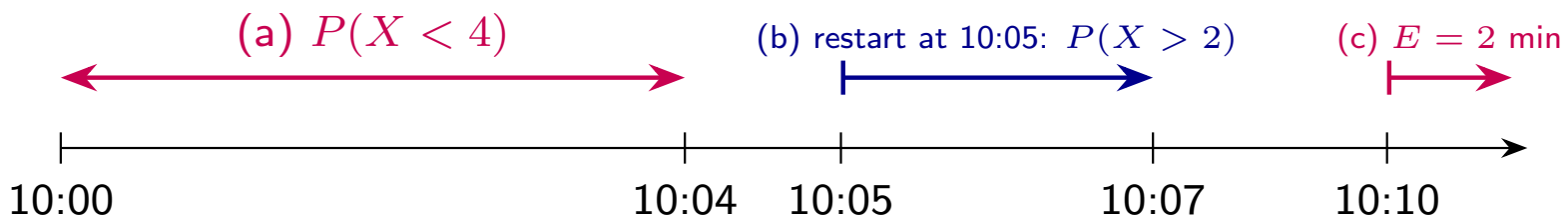
## 1. E-mails: first find $\lambda$ from the mean

$E[X] = \dfrac{1}{\lambda} = 2$ min $\Rightarrow \lambda = 0.5$ per minute. $X \equiv$ time from the last e-mail to the next one, $X \sim \exp(0.5)$.

(a) $P(X < 4) = F(4) = 1 - e^{-0.5\times 4} = 1 - e^{-2} \approx 0.865$.

(b) No e-mail until 10:05, none until 10:07? By memorylessness the 5 minutes already waited do not count:
$P(X > 7 \mid X > 5) = P(X > 2) = e^{-0.5\times 2} = e^{-1} \approx 0.368$.

(c) At 10:10 the remaining wait is again $\exp(0.5)$ with mean 2 min $\Rightarrow$ expect the next e-mail at 10:12 AM.



# Class problem 6

2. The time to weld a part is according to an exponential distribution with parameter 3 (per hour). There are 4 welding stations that begin to weld one part each.
   (a) What is the probability that the first weld would be completed in 10 minutes?
   (b) When do you expect the first weld to complete?
   (c) When do you expect the second weld to complete?
   (d) The starting time of welding the 4 parts in the respective stations are 10am, 10:05am, 10:10am and 10:15am. At 10:20am if none of the welders complete, what is the probability that the first job would be completed before 10:25am? If $Y$ denotes the time (after 10:20am) the first weld job is completed, what is the standard deviation of $Y$?

▶ Colab notebook: CP6 §2 (welding stations) `[INCLASS] [full]`

# Class problem 6

## 2. Four welding stations: the first completion is a minimum

$X_1, X_2, X_3, X_4 \sim \exp(3)$ (per hour), independent. 10 minutes $= \frac{1}{6}$ hour.

(a) First weld completed within 10 minutes $\Leftrightarrow$ $\min(X_1, X_2, X_3, X_4) < \frac{1}{6}$.
$Z \equiv \min(X_1, \ldots, X_4) \sim \exp(3+3+3+3) = \exp(12)$, so

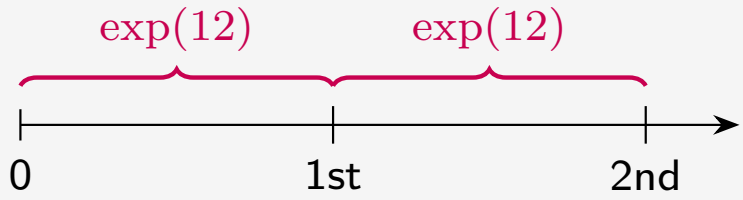
$$P\left(Z < \tfrac{1}{6}\right) = 1 - e^{-12/6} = 1 - e^{-2} \approx 0.865.$$

(b) $E[Z] = \dfrac{1}{12}$ hour $=$ 5 minutes.

▷ The rates add, not the means: four stations at rate 3 finish *something* at rate 12.

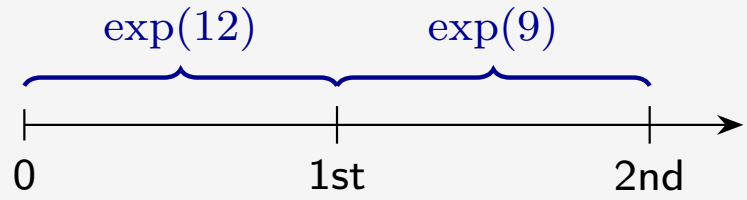
# Class problem 6

## 2(c). The second completion: two readings of the story




stations start a new part immediately after finishing one: still 4 clocks

$E = \frac{1}{12} + \frac{1}{12} = \frac{1}{6}$ h (10 min)




stations stop after finishing their part: 3 clocks remain (memoryless!)

$E = \frac{1}{12} + \frac{1}{9} = \frac{7}{36}$ h (11.7 min)

## 2(d). At 10:20 all four are still welding — the elapsed times are irrelevant

Memorylessness: each remaining welding time is again $\exp(3)$, whatever its start time. $Y \equiv$ time after 10:20 until the first completion $= \min(X_1, \ldots, X_4) \sim \exp(12)$. So $P\left(Y < \frac{1}{12}\right) = 1 - e^{-1} \approx 0.632$, and $E[Y] = \frac{1}{12}$, $\mathrm{Var}[Y] = \frac{1}{12^2} \Rightarrow \mathrm{Std}(Y) = \frac{1}{12}$ h $= 5$ min.

# Poisson process

- Consider a sequence of iid exponential random variables $X_1, X_2, \ldots$ with parameter $\lambda$. Define

$$S_n = \sum_{i=1}^{n} X_i.$$

  We can think that $S_i$ is the time of $i^{\text{th}}$ arrival.
- Let $N(t)$ be the number of arrivals until time $t$. Then,

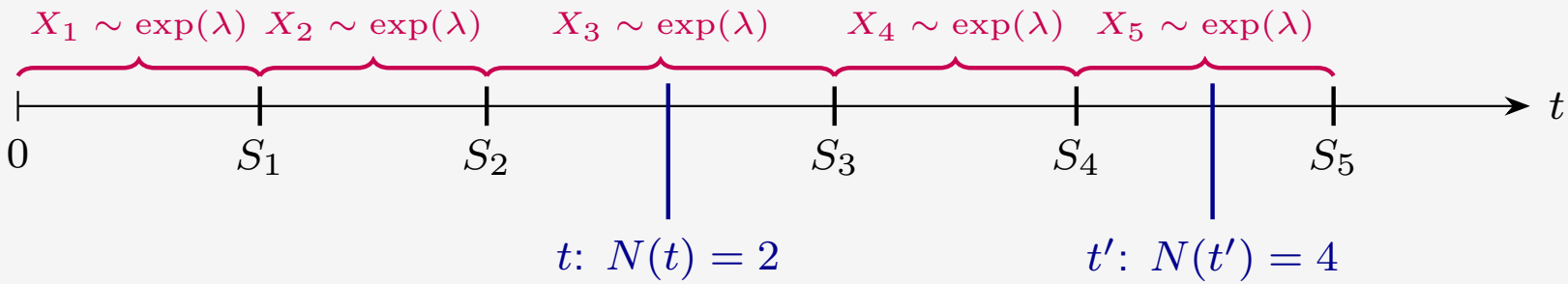
$$P\left(N(t) = k\right) = \frac{e^{-\lambda t}(\lambda t)^k}{k!}, \text{ for } k = 0, 1, 2, \ldots.$$

- The process $\{N(t), t \geq 0\}$ is called a Poisson process with parameter $\lambda$ and is denoted by $PP(\lambda)$.

▷ Companion §1.2 and Proof 2: from iid $\exp(\lambda)$ inter-arrivals to $N(t) \sim \text{Poisson}(\lambda t)$.

# Poisson process

## A counting process built from exponential gaps

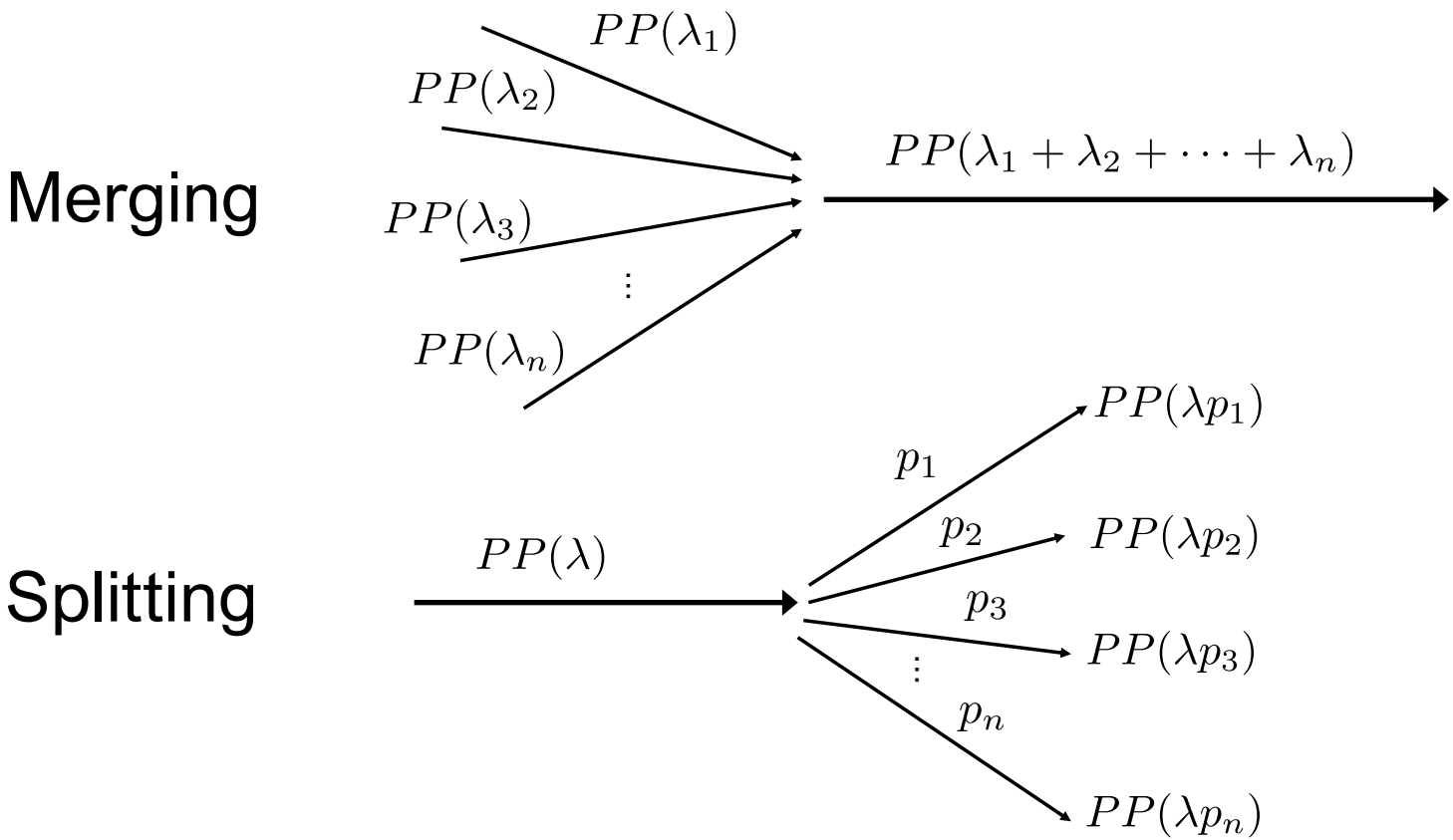



$S_n = X_1 + \cdots + X_n$ is the time of the $n$-th arrival; $N(t)$ counts the arrivals in $(0, t]$, so $N(t) \geq n \Leftrightarrow S_n \leq t$. For iid $\exp(\lambda)$ gaps the count is Poisson: $N(t) \sim$ Poisson$(\lambda t)$, hence $E[N(t)] = \lambda t$ and $\mathrm{Var}[N(t)] = \lambda t$ (Companion Proof 2).

## What "rate $\lambda$" means

$\lambda =$ mean number of arrivals per unit time $= 1/E[X_i]$. Arrivals in disjoint intervals are independent, and the number in an interval of length $t$ depends only on $t$ (independent, stationary increments) — both follow from the memoryless gaps.

# Splitting and merging




▷ Companion Proof 3: thinning and superposition. Round-robin routing gives Erlang inter-event times, not Poisson.

# Splitting and merging

**Splitting is *probabilistic*: each event goes to branch $i$ with probability $p_i$, independently of everything else**

Then branch $i$ is again a Poisson process, $PP(\lambda p_i)$, and the branches are independent (Companion Proof 3). Merging independent $PP(\lambda_i)$ gives $PP(\lambda_1 + \cdots + \lambda_n)$: the next event is the minimum of the exponential gaps.

**Round-robin routing is *not* splitting**

Send events $1, 2, \ldots, n$ to branches $1, 2, \ldots, n$ in turn, then start over. The inter-event time on one branch is then

$$X_1 + X_2 + \cdots + X_n, \qquad X_i \sim \exp(\lambda) \text{ iid} \qquad \Rightarrow \qquad \text{Erlang}(n, \lambda),$$

not exponential: the branch is a renewal process (Ch. 7) with the same mean gap $n/\lambda$ as $PP(\lambda/n)$ but a smaller variance ($n/\lambda^2$ instead of $n^2/\lambda^2$) — a more regular stream.

▷ Test it in CP6 §3: thin a $PP(10)$ with $p = 0.4$ and compare with sending every second car to the left lane.

# Class problem 6

3. Cars arrive at East Gate intersection according to a Poisson Process with average arrival rate of 10 cars per minute.
   (a) What is the probability that no cars arrive in 10 seconds?
   (b) What is the probability that more than 2 cars arrive in a 15 second interval?
   (c) What is the probability that 5 cars arrive in 2 minutes?
   (d) On an average, how many cars do you expect would arrive in 10 minutes?
   (e) What is the variance of the number of arrivals in 10 minutes?
   (f) If 40% of cars turn onto Jigok-dong and 50% turn right on Hoja market, what is the average inter-arrival time on the left lane, the center lane and the right lane?
   (g) Trucks arrive at the intersection according to a Poisson Process with average arrival rate of 8 trucks per minute and buses according to a Poisson Process with average arrival rate of 2 buses per minute. Assuming there are no other vehicles, what is the probability that 2 vehicles arrive in 6 seconds?

▶ Colab notebook: CP6 §3 (East Gate traffic) `[INCLASS] [full]`

# Class problem 6

**3. East Gate: $\lambda = 10$ cars per minute, so $N(t) \sim \text{Poisson}(10t)$ with $t$ in minutes**

(a) 10 seconds $= \frac{1}{6}$ min: $P\big(N(\frac{1}{6}) = 0\big) = \dfrac{e^{-10/6}\,(10/6)^0}{0!} = e^{-5/3} \approx 0.189$.

(b) 15 seconds $= \frac{1}{4}$ min, $\lambda t = 2.5$: $P\big(N(\frac{1}{4}) > 2\big) = 1 - P(N = 0) - P(N = 1) - P(N = 2) = 1 - e^{-2.5}\left(1 + 2.5 + \dfrac{2.5^2}{2}\right) \approx 0.456$.

(c) $P\big(N(2) = 5\big) = \dfrac{e^{-20}\,20^5}{5!} \approx 5.5 \times 10^{-5}$ — 5 cars in 2 minutes is very unlikely when 20 are expected.

(d) $E[N(10)] = \lambda t = 10 \times 10 = \boxed{100}$ cars.

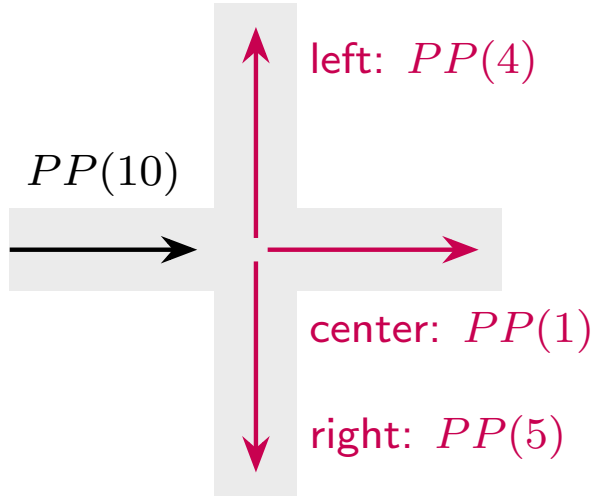
(e) $\text{Var}[N(10)] = \lambda t = 100$ (for a Poisson count, variance = mean).

▷ Always convert the time to the unit of $\lambda$ first; the Poisson parameter is $\lambda t$, never $\lambda$ alone.

# Class problem 6

## 3(f). Splitting the stream at the intersection

Each car turns left (Jigok-dong) w.p. $0.4$, right (Hoja market) w.p. $0.5$, goes straight (Posville) w.p. $0.1$, independently of the others. Thinning:
left lane $PP(10 \times 0.4) = PP(4) \Rightarrow$ mean gap $\frac{1}{4}$ min;
center lane $PP(10 \times 0.1) = PP(1) \Rightarrow 1$ min;
right lane $PP(10 \times 0.5) = PP(5) \Rightarrow \frac{1}{5}$ min.




## 3(g). Merging trucks and buses

$PP(8)$ and $PP(2)$ merge into $PP(10)$ vehicles per minute; 6 seconds $= \frac{1}{10}$ min:

$$P\big(N(\tfrac{1}{10}) = 2\big) = \frac{e^{-1}\,1^2}{2!} = \frac{e^{-1}}{2} \approx 0.184.$$

# Modeling a system as a CTMC

1. Define $X(t)$, the state at time $t$.
2. Write down $\mathcal{S}$, the state space.
3. Verify Markov & time-homogeneity properties
4. Derive $P(t)$, the transition probability matrix at time $t$
   $\Rightarrow Q \equiv$ (infinitesimal) generator matrix.
5. Draw a rate diagram.

▷ Companion §1.3 (recipe). Every CP7 problem follows these five steps; `CTMC(Q, states)` checks that rows sum to 0.

# Markov property

- $X(t) \equiv$ state of the system at time $t$
- $\mathcal{S} \equiv$ state space
- Markov property for CTMC

$$P\left(X(t+s)=j|X(s)=i, X(u)=x(u) \text{ for } 0 \leq u < s\right)$$

$$= P\left(X(t+s)=j|X(s)=i\right)$$

$$= P\left(X(t)=j|X(0)=i\right)$$

$$= p_{ij}(t).$$

- $P(t) = [p_{ij}(t)]$: transition probability matrix
  $\Rightarrow$ does not mean *one jump* from $i$ to $j$ during $t$ amount of time.

▷ Companion §1.3 and Proof 4: Markov property $\Leftrightarrow$ exponential sojourn times; the embedded chain.

# Markov property

## Two separate assumptions hide in the definition (as for a DTMC)

$$P\big(X(t+s)=j \mid X(s)=i,\ \underbrace{X(u)=x(u) \text{ for } 0\le u<s}_{\text{ignored}}\big)$$

$$\overset{(1)}{=} P(X(t+s)=j \mid X(s)=i) \overset{(2)}{=} P(X(t)=j \mid X(0)=i) =: p_{ij}(t)$$

(1) **Markov property**: the history before $s$ is irrelevant once $X(s)$ is known.
(2) **Time homogeneity**: only the elapsed time $t$ matters, not the clock time $s$.

## What the Markov property forces on the clocks

Markov property $\iff$ exponential sojourn (inter-event) times

Reason: given "still in state $i$ after $s$ time units", the remaining sojourn must not depend on $s$ — memorylessness, and the exponential is the only continuous distribution with it (Companion Proof 4). Hence: exponential clocks, one per possible transition.

# Class problem 7

1. Consider a machine that operates for an exp($\mu$) amount of time and then fails. Once it fails, it gets repaired. The repair time is an exp($\lambda$) random variable, and is independent of the past. The machine is as good as new after the repair is complete. Let $X(t)$ be the state of the machine at time $t$, 1 if it is up and 0 if it is down. Model this as a CTMC.

▶ Colab notebook: CP7 §1 (machine up/down) `[INCLASS] [full]`

# Class problem 7

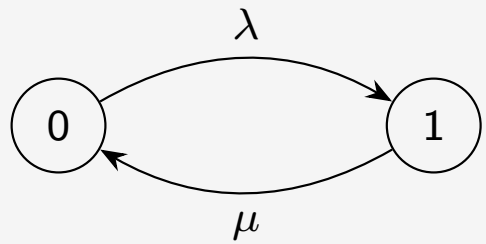
## 1. One machine: state, state space, generator

$$X(t) = \begin{cases} 1 & \text{if the machine is up at time } t \\ 0 & \text{if it is down at time } t \end{cases}, \qquad S = \{0, 1\}.$$

Up$\to$down at rate $\mu$ (up time $\exp(\mu)$);
down$\to$up at rate $\lambda$ (repair time $\exp(\lambda)$):

$$Q = \begin{array}{c|cc} & 0 & 1 \\ \hline 0 & -\lambda & \lambda \\ 1 & \mu & -\mu \end{array}$$

row sum $= 0$




rate diagram: arrows carry *rates*, no self-loops

## Why no self-loops, and why the diagonal is negative

$p_{ii}(t) = P(X(t) = i \mid X(0) = i)$ is *not* a self-transition: it includes leaving $i$ and coming back before $t$. $q_{ii} = -\sum_{j \neq i} q_{ij}$ is minus the total rate of leaving $i$, so the sojourn time in $i$ is $\exp(-q_{ii})$: here $\exp(\lambda)$ in $0$ and $\exp(\mu)$ in $1$.

# Class problem 7

2. There are two identical photocopy machines in our department. The up times of each machine is exponentially distributed with mean up time 2.5 days. As soon as a machine breaks down, a repair person is called from the company. The down times of each machine is exponentially distributed with mean half a day (down time includes the time between when a machine goes down until it is repaired and ready. It usually includes the time for the repair person to get to the department and complete the repair). It is possible that both machines are down and we have 2 repair persons from the company taking care of them. Assume that the repair persons are independent. Let $X(t)$ be the number of machines that are up at time $t$. Model the $\{X(t), t \geq 0\}$ process as a CTMC.

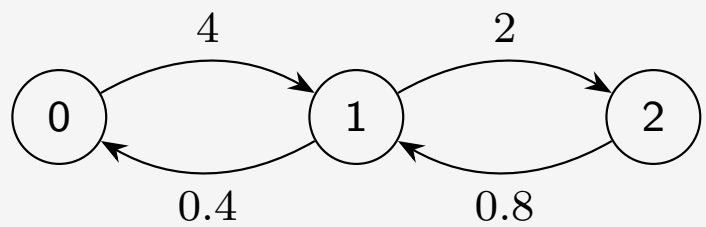
▶ Colab notebook: CP7 §2 (photocopy machines) `[INCLASS] [full]`

# Class problem 7

## 2. Two photocopy machines, two repair persons: $X(t) \equiv$ number of machines up, $S = \{0, 1, 2\}$

Up time $\sim \exp(\lambda)$ with $\frac{1}{\lambda} = 2.5$ days $\Rightarrow \lambda = 0.4$ per day; down time $\sim \exp(\mu)$ with $\frac{1}{\mu} = 0.5$ day $\Rightarrow \mu = 2$ per day.
Two machines up: each fails at rate $0.4$, and $P(X_1 = X_2) = 0$ for independent exponentials — they cannot fail at the same instant — so $2 \to 1$ at rate $2 \times 0.4$. Two down, two repair persons: $0 \to 1$ at rate $2 \times 2$.

$$Q = \begin{array}{c|ccc} & 0 & 1 & 2 \\ \hline 0 & -4 & 4 & 0 \\ 1 & 0.4 & -2.4 & 2 \\ 2 & 0 & 0.8 & -0.8 \end{array}$$




▷ Each entry is the rate of the exponential clock that moves the system: $q_{01} = 2\mu$, $q_{10} = \lambda$, $q_{12} = \mu$, $q_{21} = 2\lambda$; the diagonal makes each row sum to $0$.

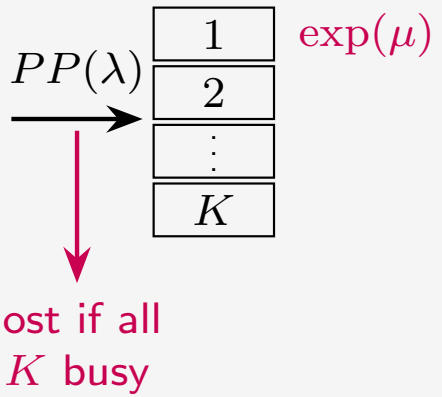
# Class problem 7

3. A telephone switch can handle a maximum of $K$ calls at any one time. Calls arrive at the switch such that the inter-arrival times of calls are according to exp($\lambda$). If the switch is already serving $K$ calls when a call arrives, then the new call is lost. If a call is accepted, it lasts for an exp($\mu$) amount of time and then terminates. All call durations are independent of each other. Let $X(t)$ be the number of calls that are being handled by the switch at time $t$. Model the $\{X(t), t \geq 0\}$ process as a CTMC.

▶ Colab notebook: CP7 §3 (telephone switch) `[INCLASS] [full]`

# Class problem 7

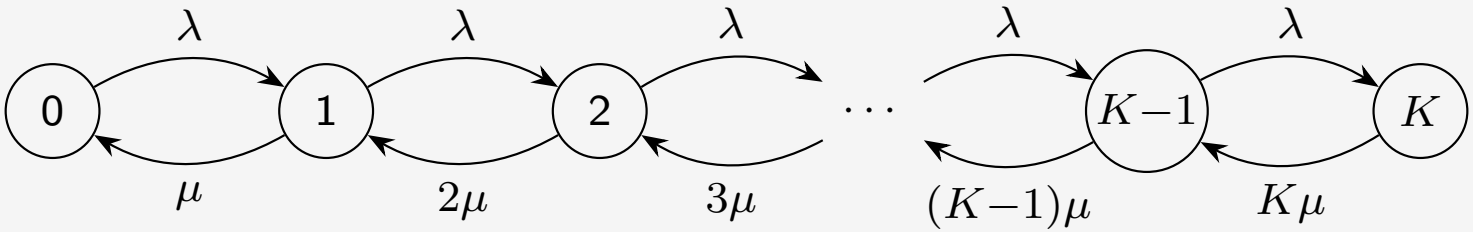
## 3. Telephone switch: $X(t) \equiv$ number of calls in the switch at time $t$, $S = \{0, 1, \ldots, K\}$




Arrivals: $i \to i+1$ at rate $\lambda$ for $i < K$ (lost when $i = K$). Departures: $i$ calls in progress $= i$ independent $\exp(\mu)$ clocks $\Rightarrow i \to i-1$ at rate $i\mu$.

$$Q = \begin{pmatrix} -\lambda & \lambda & & & \\ \mu & -(\mu+\lambda) & \lambda & & \\ & 2\mu & -(2\mu+\lambda) & \lambda & \\ & & \ddots & \ddots & \ddots \\ & & & K\mu & -K\mu \end{pmatrix}$$

## Rate diagram: a birth & death process



# Class problem 7

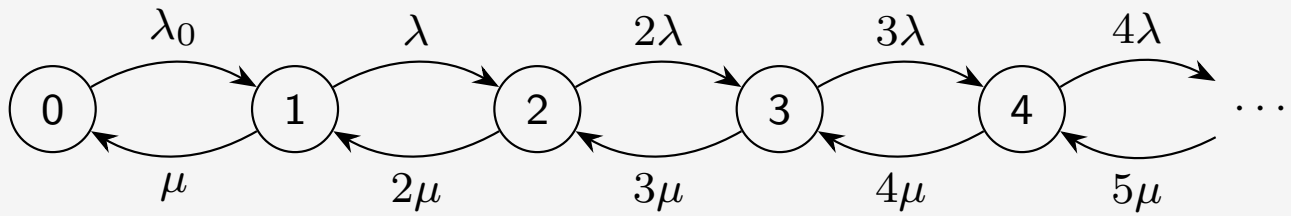
5. Consider a colony of amoebae whose lifetimes are independent exp($\mu$) random variables. During its lifetime, each amoeba produces offsprings (i.e. splits into two) in a $PP(\lambda)$ fashion. All amoebae behave independently. As soon as the last amoeba in the colony dies, a lab technician introduces a new amoeba after an exp($\lambda_0$) time. Let $X(t)$ be the number of amoebae in the colony at time $t$. Model $\{X(t), t \geq 0\}$ as a CTMC.

▶ Colab notebook: CP7 §5 (amoebae) `[INCLASS] [full]`

# Class problem 7

## 5. Amoebae: $X(t) \equiv$ number of amoebae in the colony at time $t$, $S = \{0, 1, 2, \ldots\}$

$n \geq 1$ amoebae: $n$ independent death clocks $\exp(\mu)$ and $n$ splitting clocks $\exp(\lambda) \Rightarrow n \to n-1$ at rate $n\mu$, $n \to n+1$ at rate $n\lambda$. From $0$ the technician restarts the colony: $0 \to 1$ at rate $\lambda_0$.




## Generator, read off the diagram — a birth & death process on an infinite state space

$$Q = \begin{pmatrix} -\lambda_0 & \lambda_0 & 0 & 0 & \cdots \\ \mu & -(\mu+\lambda) & \lambda & 0 & \cdots \\ 0 & 2\mu & -2(\mu+\lambda) & 2\lambda & \cdots \\ \vdots & \ddots & \ddots & \ddots & \ddots \end{pmatrix}$$

the rows were left for you to fill in class — check that each one sums to $0$

# Class problem 7

4. There are two different machines on a shopfloor with respective up times exp($\mu_1$) and exp($\mu_2$). There is a single repair person who is responsible for both these machines. The repair times are respectively exp($\lambda_1$) and exp($\lambda_2$). The repair person uses a first-come-first-serve (FCFS) policy. Model this system as a CTMC.

- Colab notebook: CP7 §4 (two machines, one repair person) `[INCLASS] [full]`

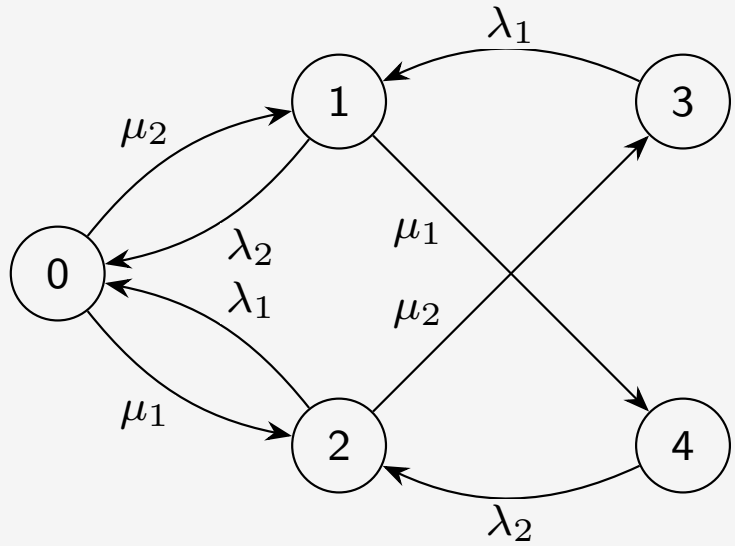
# Class problem 7

## 4. Two different machines, one repair person (FCFS): the state must record who is being repaired

$X(t) =$

$$\begin{cases} 0 & \text{both machines up} \\ 1 & \text{m/c 1 up, m/c 2 down} \\ 2 & \text{m/c 1 down, m/c 2 up} \\ 3 & \text{both down, m/c 1 in repair} \\ 4 & \text{both down, m/c 2 in repair} \end{cases}$$

$S = \{0, 1, 2, 3, 4\}$; states 3 and 4 differ only in *who failed first* — FCFS makes that matter.




## Reading the arrows

$1 \to 4$: while m/c 2 is being repaired, m/c 1 fails (rate $\mu_1$) and waits; $4 \to 2$: the repair of m/c 2 completes (rate $\lambda_2$) and the repair person turns to m/c 1. Symmetrically $2 \to 3 \to 1$. $Q$ $(5 \times 5)$: read it off the diagram.

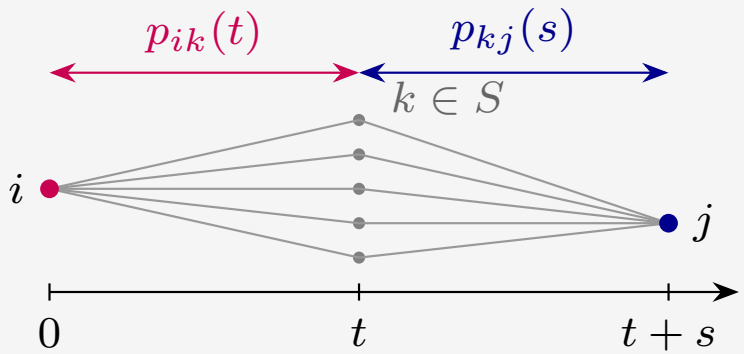
# Transition probability matrix

- $P(t) = [p_{ij}(t)]$
- $p_{ij}(t) = P\left(X(t) = j | X(0) = i\right)$
- Semigroup property: $P(t+s) = P(t)P(s)$

▷ Companion Proof 5: Chapman–Kolmogorov → Kolmogorov forward equations → $P(t) = e^{Qt}$.

# Transition probability matrix

## Semigroup property = Chapman–Kolmogorov: condition on where the chain is at time $t$




Sum over the intermediate state $k$ at time $t$; the Markov property removes $X(0) = i$ from the first factor, time homogeneity turns the interval $(t, t+s]$ into $(0, s]$. In matrix form

$$P(t+s) = P(t)P(s) .$$

$$\begin{aligned} p_{ij}(t+s) &= P(X(t+s) = j \mid X(0) = i) \\ &= \sum_{k \in S} P(X(t+s) = j,\ X(t) = k \mid X(0) = i) \\ &= \sum_{k \in S} P(X(t+s) = j \mid X(t) = k)\, P(X(t) = k \mid X(0) = i) \\ &= \sum_{k \in S} p_{kj}(s)\, p_{ik}(t) \ = \ \sum_{k \in S} p_{ik}(t)\, p_{kj}(s) \end{aligned}$$

▷ Compare Ch. 4: $P^{(m+n)} = P^{(m)} P^{(n)}$. But there is no "one step" here — differentiate instead: next frame.

# Transition probability matrix

## Differentiate the semigroup property: the Kolmogorov forward equation

$$\frac{d}{dt}P(t) = \lim_{h\to 0}\frac{P(t+h)-P(t)}{h} = \lim_{h\to 0}\frac{P(t)P(h)-P(t)}{h} \quad \text{(semigroup)}$$

$$= P(t)\lim_{h\to 0}\frac{P(h)-I}{h} = P(t)\,Q, \qquad \text{where} \quad Q \equiv \lim_{h\to 0}\frac{P(h)-I}{h}.$$

$\frac{d}{dt}P(t) = P(t)\,Q$ — $Q$ is the (infinitesimal) **generator** matrix.

## What the entries of $Q$ are

In a short interval $h$: $p_{ij}(h) = \lambda_{ij}h + o(h)$ for $j \neq i$ ($f$ is $o(h)$ if $f(h)/h \to 0$), and $p_{ii}(h) = 1 - \sum_{j\neq i} p_{ij}(h)$. Hence

$$q_{ij} = \lim_{h\to 0}\frac{p_{ij}(h)-0}{h} = \lambda_{ij} \qquad (i \neq j),$$

$$q_{ii} = \lim_{h\to 0}\frac{p_{ii}(h)-1}{h} = \lim_{h\to 0}\frac{-\sum_{j\neq i}\lambda_{ij}h + o(h)}{h} = -\sum_{j\neq i}\lambda_{ij}.$$

So $q_{ij}$ is the transition *rate* $i \to j$ and each row of $Q$ sums to $0$.

# Transition probability matrix

## Solving $\frac{d}{dt}P(t) = P(t)Q$ with $P(0) = I$

For a scalar, $x'(t) = q\,x(t)$, $x(0) = 1$ gives $x(t) = e^{qt}$. The matrix version is the same formula,

$$P(t) = e^{Qt}, \qquad \text{with} \quad e^{A} \equiv \sum_{n=0}^{\infty} \frac{A^n}{n!} = I + A + \frac{A^2}{2!} + \frac{A^3}{3!} + \cdots \quad (A^0 = I),$$

by analogy with $e^x = \sum_{n\geq 0} x^n/n!$. Check:

$$\frac{d}{dt}\sum_{n\geq 0}\frac{Q^n t^n}{n!} = \sum_{n\geq 1}\frac{Q^n t^{n-1}}{(n-1)!} = \Big(\sum_{m\geq 0}\frac{Q^m t^m}{m!}\Big)Q = P(t)\,Q, \text{ and } P(0) = I.$$

## In practice

$e^{Qt}$ is computed numerically: `scipy.linalg.expm(Q*t)`, or by integrating $\frac{d}{dt}p(t) = p(t)Q$ for the row vector $p(t) = p(0)P(t)$ with `solve_ivp`. The truncated series $\sum_{n\leq N} Q^n t^n/n!$ and $(I + Qt/N)^N$ also converge — CP8 §8 shows how large $N$ must be.

# Ergodic CTMCs

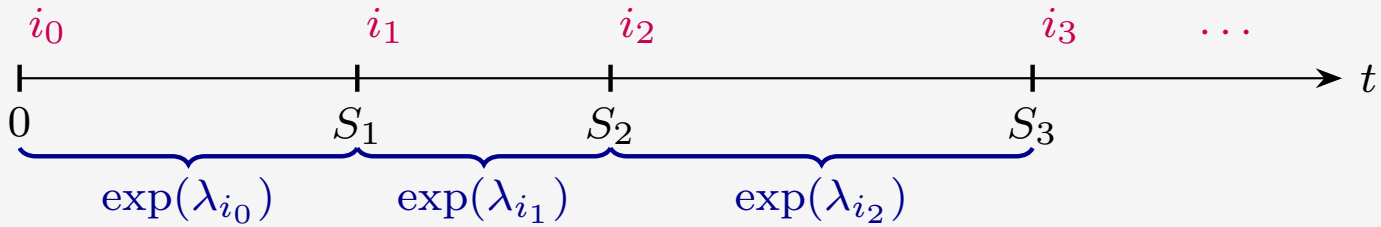
- Embedded Markov chain

| CTMC (Rate diagram) | | Embedded Markov chain (Transition diagram) |
|---|---|---|
| Irreducible | | Irreducible |
| Positive recurrent | | Positive recurrent |

▷ Companion §1.4 and Proof 4: the embedded chain; irreducibility carries over, positive recurrence need not.

# Ergodic CTMCs

## Embedded Markov chain: look at the CTMC only when it jumps




$S_n$ = time of the $n$-th jump, $X_n \equiv X(S_n)$ = state entered. $\{X_n, n \geq 0\}$ is a discrete-time Markov chain, the *embedded MC* of the CTMC. From $i$ the CTMC stays $\exp(\lambda_i)$, $\lambda_i = \sum_{j \neq i} \lambda_{ij}$, then jumps to $j$ w.p.

$$P(X_{n+1} = j \mid X_n = i) = \frac{\lambda_{ij}}{\lambda_i} = \frac{q_{ij}}{-q_{ii}} \qquad \text{(clock } i \to j \text{ rings first).}$$

## Which clock rings first? $X \sim \exp(\alpha)$, $Y \sim \exp(\beta)$ independent

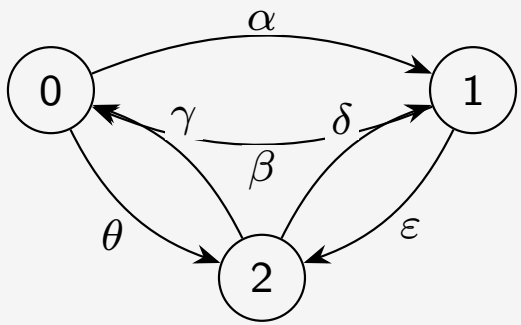
$$P(X < Y) = \int_0^\infty \int_0^y f_X(x) f_Y(y)\, dx\, dy = \int_0^\infty (1 - e^{-\alpha y})\, \beta e^{-\beta y} dy = \frac{\alpha}{\alpha + \beta};$$

three clocks: $P(X < \min(Y, Z)) = \dfrac{\alpha}{\alpha + (\beta + \gamma)}$, as $\min(Y, Z) \sim \exp(\beta + \gamma)$.

# Ergodic CTMCs

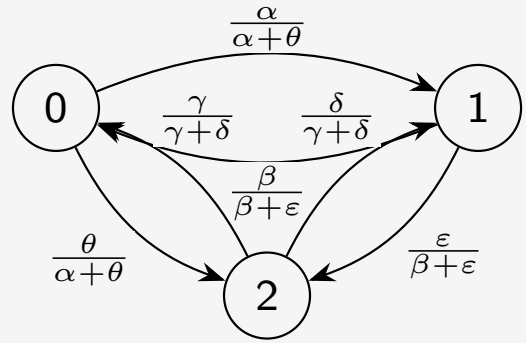
## Example: a CTMC and its embedded chain

rate diagram (CTMC)




sojourn in 0: $\exp(\alpha+\theta)$

transition diagram (embedded MC)




no self-loops; rows sum to 1

The embedded chain keeps the *arrows* and forgets the *speeds*: $\pi = \pi P$ counts jumps, not time.

## Which properties carry over?

- Irreducible: $\exists\, t > 0$: $p_{ij}(t) > 0\ \forall i, j \iff \exists\, n > 0$: $p_{ij}^{(n)} > 0\ \forall i, j$ (same arrows) — yes.
- Positive recurrent: *not* equivalent in general. With $\pi$ the stationary distribution of the embedded chain, the CTMC is positive recurrent iff $\sum_j \pi_j/\lambda_j < \infty$ (finite $S$: automatic).

# Ergodic CTMCs

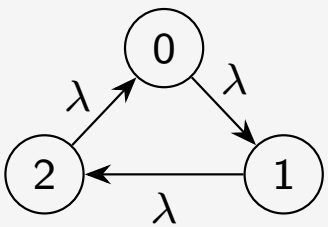
- Aperiodic?

▷ Companion §1.4: no aperiodicity condition in continuous time — sojourn times are continuous, so nothing can cycle.

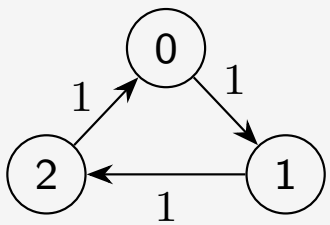
# Ergodic CTMCs

## No aperiodicity condition in continuous time

The system stays in state $i$ for an $\exp(\lambda_i)$, $\lambda_i = \sum_{j \in S} \lambda_{ij}$, amount of time — a continuous random variable, so a return to $i$ can happen at any time, never only at multiples of some $d$: no state can be periodic.




CTMC




embedded: period 3

the embedded chain cycles, but $P(X(t) = 0) \to \frac{1}{3}$ smoothly

**Ergodic CTMC = irreducible + positive recurrent.** Then

$$p_j \equiv \lim_{t \to \infty} p_{ij}(t) = \lim_{t \to \infty} P(X(t) = j)$$

exists and does not depend on the start $i$.

## $p_j$ (CTMC) versus $\pi_j$ (embedded MC): equal? In general no!

$\pi_j$ counts *visits* (each jump weighs the same); $p_j$ is the fraction of *time* in $j$. A random observer more often lands in a long sojourn, so states with small $\lambda_j$ weigh more in $p$ than in $\pi$: CP8 §1 gives $p = (\frac{\mu}{\lambda+\mu}, \frac{\lambda}{\lambda+\mu})$ but $\pi = (\frac{1}{2}, \frac{1}{2})$.

# Limiting probability

- Kolmogorov forward equation.

$$\frac{d}{dt}P(t) = P(t)Q.$$

- In steady state, $P(\infty)$ will not change.

$$\lim_{t\to\infty} P(t)Q = P(\infty)Q = 0$$

▷ Companion §1.5 and Proof 6: $pQ = 0$ is node balance (rate out = rate in); $p$ versus the embedded $\pi$.

# Limiting probability

## From $P(\infty)Q = 0$ to a linear system for $\boldsymbol{p}$

Ch. 4: $P^n \to \begin{pmatrix} \boldsymbol{\pi} \\ \boldsymbol{\pi} \\ \vdots \end{pmatrix}$ — every row is $\boldsymbol{\pi}$.

Ch. 5:

$$P(t) = e^{Qt} \to P(\infty) = \begin{pmatrix} \boldsymbol{p} \\ \boldsymbol{p} \\ \vdots \end{pmatrix},$$

$\boldsymbol{p} = (p_0, p_1, \ldots)$: every row is the same because the start is forgotten.

$\frac{d}{dt}P(t) = P(t)Q$ and $P(t)$ stops changing, so $P(\infty)Q = 0$. Every row of $P(\infty)$ is $\boldsymbol{p}$, so this is one equation:

$$\boldsymbol{p}Q = 0, \qquad \sum_{i \in S} p_i = 1 \,.$$

$|S|$ unknowns. The $|S|$ equations of $\boldsymbol{p}Q = 0$ are the columns of $Q$, and these sum to the zero vector (every row of $Q$ sums to $0$): one equation is redundant and the normalisation replaces it.

▷ Solving $\boldsymbol{p}Q = 0$: expand and solve (CP8 §1), node balance (CP8 §2), or arc balance for birth & death processes (CP8 §3).

# Class problem 8

1. Consider a machine that operates for an exp($\mu$) amount of time and then fails. The repair time is an exp($\lambda$) random variable. What is the probability that the machine is up when it is observed in steady state?

▶ Colab notebook: CP8 §1 (machine, steady state) `[INCLASS] [full]`

# Class problem 8

## 1. Machine up/down in the long run: solve $\boldsymbol{pQ} = 0$, $p_0 + p_1 = 1$

$$\boldsymbol{pQ} = (p_0 \;\; p_1)\begin{pmatrix} -\lambda & \lambda \\ \mu & -\mu \end{pmatrix} = (0 \;\; 0) \;\Rightarrow\; -\lambda p_0 + \mu p_1 = 0, \;\; \lambda p_0 - \mu p_1 = 0$$

(the same equation twice: one equation of $\boldsymbol{pQ} = 0$ is always redundant).

So $p_1 = \dfrac{\lambda}{\mu} p_0$, and with $p_0 + p_1 = 1$: $\quad p_0 = \dfrac{\mu}{\lambda + \mu}, \quad p_1 = \dfrac{\lambda}{\lambda + \mu}$

($p_1 =$ availability).

## Contrast: the embedded chain

$P = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \Rightarrow \pi_0 = \pi_1 = \frac{1}{2}$: half of the *jumps* land in each state, but the *time* splits as $\mu : \lambda$ — $p_j \propto \pi_j / \lambda_j$ with $\lambda_0 = \lambda$, $\lambda_1 = \mu$ (Companion Proof 6).

▷ Sanity check: up $1/\mu$, down $1/\lambda$ per cycle $\Rightarrow$ fraction up $= \dfrac{1/\mu}{1/\mu + 1/\lambda} = \dfrac{\lambda}{\lambda + \mu}$.

# Class problem 8

2. There are two identical photocopy machines in our department. The up times of each machine is exponentially distributed with mean up time 2.5 days. The down times of each machine is exponentially distributed with mean half a day. There are 2 repair persons. What fraction of time in the long run are zero, one and two machines in working condition?

- Colab notebook: CP8 §2 (photocopy, steady state) `[INCLASS] [full]`

# Class problem 8

## 2. Two photocopy machines: $pQ = 0$ written out

$$(p_0 \;\; p_1 \;\; p_2)\begin{pmatrix} -4 & 4 & 0 \\ 0.4 & -2.4 & 2 \\ 0 & 0.8 & -0.8 \end{pmatrix} = (0 \;\; 0 \;\; 0)$$

$$\Rightarrow \quad -4p_0 + 0.4p_1 = 0, \quad 4p_0 - 2.4p_1 + 0.8p_2 = 0, \quad 2p_1 - 0.8p_2 = 0$$

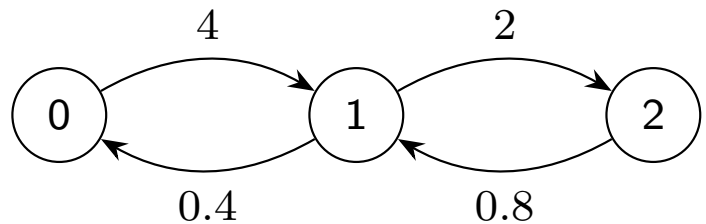
$$4p_0 = 0.4p_1\,, \qquad 2.4p_1 = 4p_0 + 0.8p_2\,, \qquad 0.8p_2 = 2p_1$$

Rearranged (last line), each equation reads rate *out of* the state $\times$ its probability $=$ rate *into* it from each neighbour $\times$ that neighbour's probability — the **node balance equations**, one per state.

## Solve

$p_1 = 10p_0, \;\; p_2 = 2.5p_1 = 25p_0;$
$p_0 + p_1 + p_2 = 1 \Rightarrow$

$$p_0 = \frac{0.1}{3.6} = \frac{1}{36}, \quad p_1 = \frac{1}{3.6} = \frac{10}{36},$$
$$p_2 = \frac{2.5}{3.6} = \frac{25}{36}.$$




$\boldsymbol{p} \approx (0.028, 0.278, 0.694)$. Check: each machine is up $\frac{2.5}{2.5+0.5} = \frac{5}{6}$ of the time, independently, so $p_2 = (\frac{5}{6})^2$.

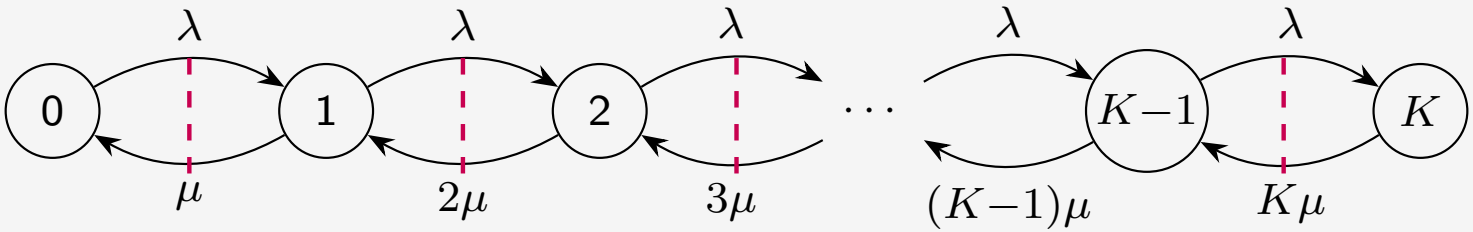
# Class problem 8

3. A telephone switch can handle a maximum of $K$ calls at any one time. Calls arrive at the switch according to $PP(\lambda)$. If the switch is already serving $K$ calls when a call arrives, then the new call is lost. If a call is accepted, it lasts for an exp($\mu$) amount of time and then terminates. What is the limiting distribution of the number of calls at the switch?

▶ Colab notebook: CP8 §3 (switch, limiting) `[INCLASS] [full]`

# Class problem 8

## 3. Birth & death: balance across each *arc* instead of each node




Cut between $k-1$ and $k$ (dashed): in the long run the flow to the right equals the flow to the left — the arc balance (detailed balance) equations,

$$\lambda p_0 = \mu p_1, \quad \lambda p_1 = 2\mu p_2, \quad \lambda p_2 = 3\mu p_3, \quad \ldots, \quad \lambda p_{K-1} = K\mu\, p_K.$$

## Solve by recursion

$$p_1 = \frac{\lambda}{\mu}p_0, \;\; p_2 = \frac{\lambda}{2\mu}p_1 = \frac{1}{2}\left(\frac{\lambda}{\mu}\right)^2 p_0, \;\; p_3 = \frac{\lambda}{3\mu}p_2 = \frac{1}{3!}\left(\frac{\lambda}{\mu}\right)^3 p_0, \;\; \ldots,$$

$$p_k = \frac{1}{k!}\left(\frac{\lambda}{\mu}\right)^k p_0; \;\; \text{then} \sum_{i=0}^{K} p_i = p_0 \sum_{i=0}^{K} \frac{1}{i!}\left(\frac{\lambda}{\mu}\right)^i = 1 \text{ fixes } p_0.$$

# Class problem 8

## Three ways to solve $\boldsymbol{p}Q = 0$, $\sum_i p_i = 1$

1. **Just expand and solve** — $|S|$ linear equations (CP8 §1, §8; `CTMC.stationary()` does exactly this).
2. **Node balance equations** — for each state $j$: (rate out of $j$)$\times p_j = \sum_{i \neq j} p_i\, q_{ij}$ (CP8 §2, §4).
3. **Arc balance equations** — the flow across every cut is balanced: $\lambda_k p_k = \mu_{k+1} p_{k+1}$ (CP8 §3, §9).

1 and 2 always work for an ergodic CTMC (they are the same equations, rearranged). 3 is valid for **birth & death processes** — more generally whenever the chain is *time-reversible* (Ch. 8: $M/M/s$ queues and product-form solutions).

▷ Arc balance is stronger than node balance: it asks $p_i q_{ij} = p_j q_{ji}$ for *every* pair. On a line it follows from node balance by induction from the left end; on a general graph it need not hold.

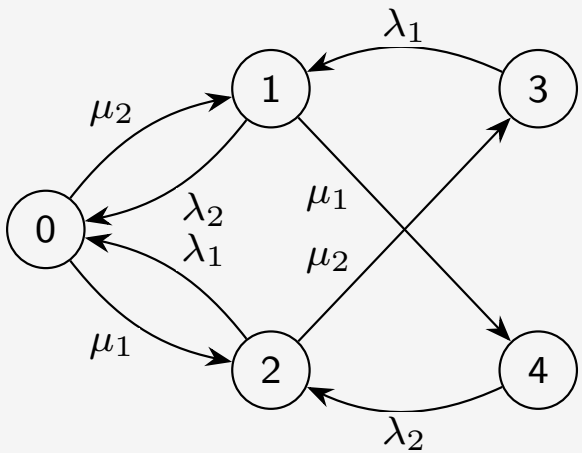
# Class problem 8

4. There are two different machines on a shopfloor with respective up times exp($\mu_1$) and exp($\mu_2$). There is a single repair person who is responsible for both these machines. The repair times are respectively exp($\lambda_1$) and exp($\lambda_2$). What fraction of time is each of the machines up?

- Colab notebook: CP8 §4 (two machines, steady state) `[INCLASS] [full]`

# Class problem 8

## 4. Two machines, one repair person: (rate out)$\times p_j$ = (rates in)$\times p_{\text{neighbour}}$, one equation per state




$$
\begin{aligned}
(\mu_1+\mu_2)\,p_0 &= \lambda_2 p_1 + \lambda_1 p_2 && (1)\\
(\lambda_2+\mu_1)\,p_1 &= \mu_2 p_0 + \lambda_1 p_3 && (2)\\
(\lambda_1+\mu_2)\,p_2 &= \mu_1 p_0 + \lambda_2 p_4 && (3)\\
\lambda_1\,p_3 &= \mu_2 p_2 && (4)\\
\lambda_2\,p_4 &= \mu_1 p_1 && (5)\\
\textstyle\sum_i p_i &= 1
\end{aligned}
$$

## Eliminate: everything in terms of $p_0$

(4)→(2): $(\lambda_2+\mu_1)p_1 = \mu_2 p_0 + \mu_2 p_2 \;\Rightarrow\; p_2 = \dfrac{\lambda_2+\mu_1}{\mu_2}\,p_1 - p_0$ (6)

(6)→(1): $(\mu_1+\mu_2)p_0 = \lambda_2 p_1 + \lambda_1 \dfrac{\lambda_2+\mu_1}{\mu_2} p_1 - \lambda_1 p_0 \;\Rightarrow$

$(\lambda_1+\mu_1+\mu_2)p_0 = \dfrac{\lambda_2\mu_2+\lambda_1\lambda_2+\lambda_1\mu_1}{\mu_2}\,p_1$ (7)

# Class problem 8

## 4, continued: the solution, with $D \equiv \lambda_1\mu_1 + \lambda_1\lambda_2 + \lambda_2\mu_2$

(7): $p_1 = \dfrac{\mu_2(\lambda_1+\mu_1+\mu_2)}{D}\,p_0$. (7)→(2):

$\dfrac{(\lambda_2+\mu_1)\mu_2(\lambda_1+\mu_1+\mu_2)}{D}p_0 = \mu_2 p_0 + \lambda_1 p_3$, so

$$p_3 = \frac{\mu_2(\lambda_2\mu_1+\mu_1^2+\mu_1\mu_2)}{\lambda_1 D}\,p_0 = \frac{\mu_1\mu_2(\lambda_2+\mu_1+\mu_2)}{\lambda_1 D}\,p_0 \quad (8).$$

(8)→(4) and (7)→(5):

$$p_2 = \frac{\mu_1(\lambda_2+\mu_1+\mu_2)}{D}\,p_0, \qquad p_4 = \frac{\mu_1\mu_2(\mu_1+\mu_2+\lambda_1)}{\lambda_2 D}\,p_0,$$

and $\sum_i p_i = 1$ fixes $p_0$. m/c 1 up: $p_0 + p_1$, m/c 2 up: $p_0 + p_2$.

## Check with numbers (CP8 §4)

$\mu_1 = 1,\ \mu_2 = 2,\ \lambda_1 = 3,\ \lambda_2 = 4$ per day:
$\boldsymbol{p} \approx (0.463,\ 0.242,\ 0.141,\ 0.094,\ 0.060)$; m/c 1 up $70.5\%$, m/c 2 up $60.4\%$ of the time.

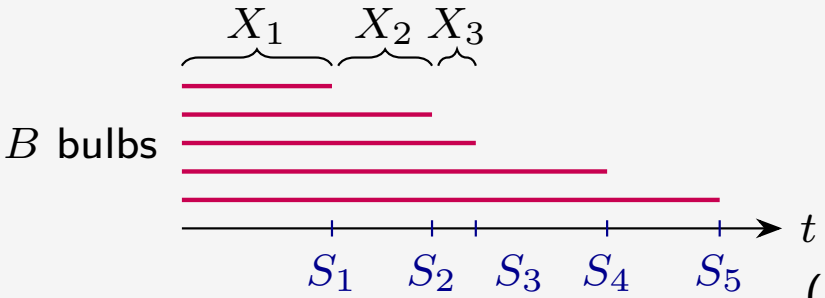
# Class problem 8

5. There are $B$ light bulbs and they are all switched on simultaneously. The lifetime of each light bulb is exponentially distributed with mean $\theta$ hours. Assume $S_0 = 0$ and for $i \geq 1$, $S_i$ denotes the time of $i$th failure (i.e. the $i$th time a light bulb dies). Define the counting process $\{N(t), t \geq 0\}$ such that $N(t) = \max\{n \geq 0 : S_n \leq t\}$ for $t \geq 0$. Answer the following questions assuming that $B > 2$:
   (a) $\{N(t), t \geq 0\}$ is a Poisson process. TRUE or FALSE?
   (b) What is $\mathsf{E}[S_2 - S_1]$?

▶ Colab notebook: CP8 §5 (light bulbs) `[INCLASS] [full]`

# Class problem 8

## 5. $B$ bulbs switched on together, each lifetime $\exp(1/\theta)$




$X_1 \sim \exp(\frac{B}{\theta})$, $X_2 \sim \exp(\frac{B-1}{\theta})$,
$X_3 \sim \exp(\frac{B-2}{\theta})$, ...

(a) **FALSE.** After $k$ failures only $B-k$ bulbs burn, so the next gap is $\exp\left(\frac{B-k}{\theta}\right)$: independent but *not identically distributed* gaps (the rate drops; after $B$ failures nothing happens). A Poisson process needs iid $\exp(\lambda)$ gaps.

(b) Between $S_1$ and $S_2$, $B-1$ bulbs are alive, each with residual life $\exp(1/\theta)$ (memoryless); their minimum is $\exp\left(\frac{B-1}{\theta}\right)$:

$$E[S_2 - S_1] = \frac{\theta}{B-1}.$$

▷ A pure death process: $B \to B-1 \to \cdots \to 0$ at rates $B/\theta, \ldots, 1/\theta$.

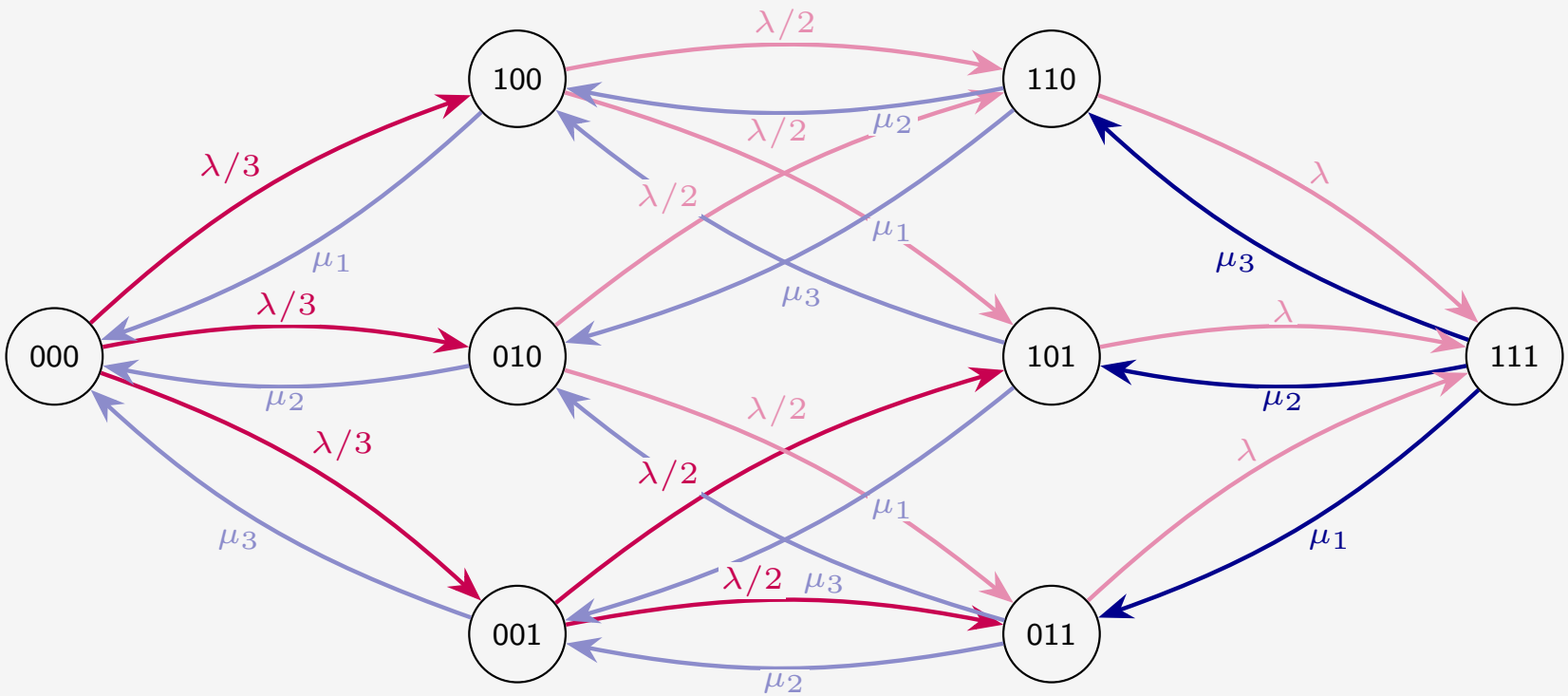
# Class problem 8

6. A barbershop has 3 servers (i.e. hair stylists) and no waiting room. Arrivals to the shop are $PP(\lambda)$. Server $i$ (for $i = 1, 2, 3$) takes exp($\mu_i$) amount of time to serve a customer. Let $(X_1(t), X_2(t), X_3(t))$ be the state of the system at time $t$, where $X_i(t)$ is the number of customers at server $i$. Assume that if there are more than one servers free, the arriving customer picks one of them at random with equal probabilities. The system can thus be modeled as an 8-state CTMC. Draw the rate diagram.

▶ Colab notebook: CP8 §6 (barbershop, rate diagram) `[INCLASS] [full]`

# Class problem 8

## 6. Barbershop with 3 stylists: state $x_1x_2x_3$ ($x_i = 1$ if stylist $i$ is busy), 8 states




arrivals split equally among the free stylists ($\lambda/3$, $\lambda/2$, $\lambda$; lost from $111$); departures at the busy stylist's own rate $\mu_i$. The lighter arcs were left to complete after class — check each one (CP8 §6 builds the $8 \times 8$ generator).

# Class problem 8

7. For the shop in the previous question, the expected service times are 20, 15, and 10 minutes. The servers have been busy for 5, 7 and 6 minutes respectively with their current customers. What is the expected remaining time until the next service completion?

▶ Colab notebook: CP8 §7 (barbershop, next completion) `[INCLASS] [full]`

# Class problem 8

## 7. Expected remaining time until the next service completion

Rates: $\mu_1 = \frac{1}{20}$, $\mu_2 = \frac{1}{15}$, $\mu_3 = \frac{1}{10}$ per minute. The stylists have been busy for 5, 7 and 6 minutes — by memorylessness these elapsed times are irrelevant: each remaining service time is again $\exp(\mu_i)$.

$$R \equiv \min(R_1, R_2, R_3) \sim \exp(\mu_1 + \mu_2 + \mu_3) = \exp\left(\tfrac{1}{20} + \tfrac{1}{15} + \tfrac{1}{10}\right) = \exp\left(\tfrac{13}{60}\right),$$

so $E[R] = \dfrac{60}{13} \approx 4.62$ minutes — the same answer whether the stylists have been busy for 5, 7, 6 minutes or for an hour.

## Two follow-ups (CP8 §7)

Which stylist finishes first? $P(\text{stylist } i \text{ first}) = \dfrac{\mu_i}{\mu_1 + \mu_2 + \mu_3}$: $\frac{3}{13}$, $\frac{4}{13}$, $\frac{6}{13}$, independently of *when* the completion happens. If the service times were *deterministic* (20, 15, 10 minutes), the next completion would be in $\min(15, 8, 4) = 4$ minutes for sure — the exponential assumption is what makes the elapsed times drop out (HW 3 Part C).

# Class problem 8

8. $\{X(t), t \geq 0\}$ is a CTMC with state space $S = \{0, 1, 2\}$ and infinitesimal generator matrix:

$$Q = \begin{pmatrix} __ & 2 & 3 \\ 1 & -1 & __ \\ __ & 0 & -2 \end{pmatrix}.$$

Fill up the blanks in $Q$. If the CTMC is in state 0 at time 0, what is the probability that the CTMC would be in state 2 at time $t = 1$? Given that $p_0 = 2/9$, compute $p_1$ and $p_2$, where $p_0$, $p_1$ and $p_2$ form the limiting distribution of the CTMC. If the cost incurred per unit time in state $i$ is $1,800$ won, what is the long run average cost incurred by the system?

- Colab notebook: CP8 §8 (Q with blanks) `[INCLASS] [full]`

# Class problem 8

## 8. Fill the blanks (rows sum to 0), then three questions

$$Q = \begin{pmatrix} -5 & 2 & 3 \\ 1 & -1 & 0 \\ 2 & 0 & -2 \end{pmatrix}.$$ **Transient:**

$P(X(1) = 2 \mid X(0) = 0) = [P(1)]_{02} = [e^Q]_{02} = \left[(1 \;\; 0 \;\; 0)\, e^Q\right]_2 \approx 0.371$, where

$$e^Q = \sum_{n=0}^{\infty} \frac{Q^n}{n!} = \lim_{n\to\infty} \left(I + \frac{Q}{n}\right)^n \approx \sum_{n=0}^{N} \frac{Q^n}{n!} \text{ or } \left(I + \frac{Q}{N}\right)^N$$ for a large $N$ (as $e^x = \sum_n x^n/n! = \lim_n (1 + x/n)^n$). Stop when one more term no longer changes the sum:

$N^* = \inf\left\{N : \left\| \sum_{n\le N} \frac{Q^n}{n!} - \sum_{n\le N-1} \frac{Q^n}{n!} \right\| < \varepsilon\right\}$ ($N^* = 34$ for $\varepsilon = 10^{-10}$).

▷ CP8 §8 compares the truncated series, $(I + Q/N)^N$, `expm` and `solve_ivp` on this $Q$, and plots $p(t) = p(0)e^{Qt}$ from each starting state.

# Class problem 8

## 8, continued: limiting distribution and cost

$\boldsymbol{p}Q = 0$ with $Q = \begin{pmatrix} -5 & 2 & 3 \\ 1 & -1 & 0 \\ 2 & 0 & -2 \end{pmatrix}$: the equation of state 1 (second column) gives $2p_0 - p_1 = 0 \Rightarrow p_1 = 2p_0$; state 2 gives $3p_0 - 2p_2 = 0 \Rightarrow p_2 = \frac{3}{2}p_0$; with $p_0 + p_1 + p_2 = 1$: $\boldsymbol{p} = \left(\frac{2}{9}, \frac{4}{9}, \frac{3}{9}\right) \approx (0.222, 0.444, 0.333)$.

**Cost:** 1,800 won per unit time *in every state*, so $C = \sum_j c(j)p_j = 1{,}800 \sum_j p_j =$ 1,800 won per unit time — $\boldsymbol{p}$ is not even needed.

## Contrast with the embedded chain

$P = \begin{pmatrix} 0 & \frac{2}{5} & \frac{3}{5} \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}$ gives $\pi = (\frac{1}{2}, \frac{1}{5}, \frac{3}{10})$: state 0 receives half of the *jumps* but only $\frac{2}{9}$ of the *time*, because its sojourn ($\exp(5)$) is the shortest. Check: $p_j \propto \pi_j/\lambda_j = (\frac{1/2}{5}, \frac{1/5}{1}, \frac{3/10}{2}) = (0.1, 0.2, 0.15) \propto (2, 4, 3)$. ✓

# Cost & rewards

- $p_j$: limiting distribution
- $c(j)$: average cost per unit time when the CTMC is in state $j$.
- $C$: long-run average cost per unit time

$$C = \sum_{j \in \mathcal{S}} c(j) p_j.$$

▷ Companion Proof 8: long-run cost per unit time $\sum_j c(j)p_j$ — time fractions, as in Ch. 4.

# Class problem 8

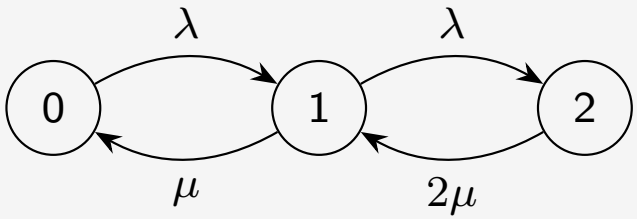
9. Consider 2 identical machines with up times and repair times as described in problem #1 above. Would it better to employ 1 or 2 repair persons for this system? If a machine is up, it generates a revenue of $r$ won per unit time. However each repair person charges $c$ won per unit time.

- Colab notebook: CP8 §9 (1 or 2 repair persons) `[INCLASS] [full]`

# Class problem 8

## 9. Two identical machines: up time $\exp(\mu)$, repair time $\exp(\lambda)$; $X(t) \equiv$ number of machines up

**Case 1: one repair person.**




Arc balance: $\lambda p_0 = \mu p_1$, $\lambda p_1 = 2\mu p_2$,
$\sum_i p_i = 1 \Rightarrow p_1 = \frac{\lambda}{\mu} p_0$, $p_2 = \frac{\lambda^2}{2\mu^2} p_0$:

$$p_0 = \frac{2\mu^2}{2\mu^2 + 2\lambda\mu + \lambda^2}, \quad p_1 = \frac{2\lambda\mu}{2\mu^2 + 2\lambda\mu + \lambda^2}, \quad p_2 = \frac{\lambda^2}{2\mu^2 + 2\lambda\mu + \lambda^2}.$$

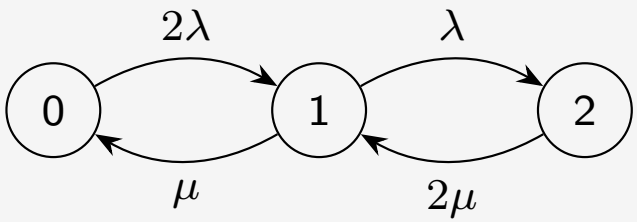
## Average reward per unit time (case 1)

Revenue $r$ per up machine; the repair person charges $c$ while working: reward $-c$ in state $0$, $r - c$ in state $1$, $2r$ in state $2$.

$$\text{reward}_1 = -c\,p_0 + (r - c)\,p_1 + 2r\,p_2 = \frac{-2\mu^2 c + 2\lambda\mu(r - c) + 2r\lambda^2}{2\mu^2 + 2\lambda\mu + \lambda^2}.$$

# Class problem 8

## Case 2: two repair persons




Arc balance: $2\lambda p_0 = \mu p_1$, $\lambda p_1 = 2\mu p_2 \Rightarrow$ $p_1 = \frac{2\lambda}{\mu} p_0$, $p_2 = \frac{\lambda^2}{\mu^2} p_0$:

$$p_0 = \frac{\mu^2}{\mu^2 + 2\lambda\mu + \lambda^2}, \qquad p_1 = \frac{2\lambda\mu}{(\mu+\lambda)^2}, \qquad p_2 = \frac{\lambda^2}{(\mu+\lambda)^2}$$

(each machine is up independently w.p. $\frac{\lambda}{\lambda+\mu}$: a binomial). Both repair persons charge while working: reward $-2c$ in state $0$, $r-c$ in state $1$, $2r$ in state $2$:

$$\text{reward}_2 = -2c\,p_0 + (r-c)\,p_1 + 2r\,p_2 = \frac{-2\mu^2 c + 2\lambda\mu(r-c) + 2r\lambda^2}{\mu^2 + 2\lambda\mu + \lambda^2}.$$

# Class problem 8

## Case 1 versus case 2: the same numerator, different denominators

$$\text{reward}_1 = \frac{N}{2\mu^2 + 2\lambda\mu + \lambda^2}, \qquad \text{reward}_2 = \frac{N}{\mu^2 + 2\lambda\mu + \lambda^2},$$

with the second denominator smaller, and the common numerator factors: $N = -2\mu^2 c + 2\lambda\mu(r-c) + 2r\lambda^2 = 2\lambda\mu(\mu+\lambda)\left(\frac{r}{\mu} - \frac{c}{\lambda}\right)$. Hence

$$\frac{r}{\mu} > \frac{c}{\lambda} \;\Rightarrow\; \text{two repair persons (case 2)}; \qquad \frac{r}{\mu} < \frac{c}{\lambda} \;\Rightarrow\; \text{one (case 1)}.$$

Reading: $r/\mu$ = revenue earned during one up period, $c/\lambda$ = cost of one repair. When a repair pays for itself, hire the second person.

▷ CP8 §9: the "repair-person dashboard" — sliders for $\lambda, \mu, r, c$ show the two rewards crossing exactly at $r/\mu = c/\lambda$.